

Lab session 3: Programming Fundamentals

Problem 1: Polynomial normal form

A polynomial in x is a sum of terms of the form ax^k , where a is an integer coefficient and k is a non-negative integer.

A polynomial is in *normal form* if terms with the same power of x have been combined and the remaining terms are ordered by decreasing exponent.

Write a C program that reads a polynomial with integer coefficients and a maximum degree of 100, and prints the polynomial in normal form. Both the input and the output contain no spaces. Both the input and the output will not contain consecutive $+$ or $-$ signs, such as $++$, $+-$, $-+$, or $--$. Terms whose coefficient is 0 must be omitted from the output. Moreover, a coefficient of 1 should not be printed explicitly for terms containing x ; for example, print x^2 instead of $1x^2$.

For example, given input $5x^{30}+2x^2-10+7x^{30}$ the output should be $12x^{30}+2x^2-10$, because the two terms containing x^{30} are combined.

Example 1:

input:

$5x^{30}+2x^2-10+7x^{30}$

output:

$12x^{30}+2x^2-10$

Example 2:

input:

42

output:

42

Example 3:

input:

$5x^{30}+2x^2-10-7x^{30}+52$

output:

$-2x^{30}+2x^2+42$

Problem 2: Polynomial multiplication

Write a C program that multiplies two polynomials. Each polynomial has integer coefficients and a maximum degree of 100.

The input consists of the two polynomials enclosed in parentheses and separated by an asterisk. The input contains no spaces. Print their product in normal form, using the same output rules as in the previous exercise.

For example, given the input $(x+1) * (x+1)$ the program should produce x^2+2x+1 because $(x+1)(x+1) = x^2 + 2x + 1$.

Example 1:

input:

$(x+1) * (x+1)$

output:

x^2+2x+1

Example 2:

input:

$(x-1) * (x+1)$

output:

x^2-1

Example 3:

input:

$(5x^{30}+2x^2-10+7x^{30}) * (x+1)$

output:

$12x^{31}+12x^{30}+2x^3+2x^2-10x-10$

Problem 3: Polynomial division

Write a C program that divides one polynomial by another. Both polynomials have integer coefficients and a maximum degree of 100.

The input consists of two polynomials enclosed in parentheses and separated by a slash. The input contains no spaces. If the division has no remainder, print the quotient in normal form, using the same output rules as in the previous exercises. If the division has a non-zero remainder, the output should be `INDIVISIBLE`.

For example, given the input $(x^3 - 2x^2 + 6x - 5) / (x - 1)$ the output should be $x^2 - x + 5$ because $(x - 1)(x^2 - x + 5) = x^3 - 2x^2 + 6x - 5$.

Example 1:

input:

$(x^3 - 2x^2 + 6x - 5) / (x - 1)$

output:

$x^2 - x + 5$

Example 2:

input:

$(6x^3 + 18x^2 + 18x + 6) / (3x + 3)$

output:

$2x^2 + 4x + 2$

Example 3:

input:

$(3x^2 + 2x + 1) / (2x + 1)$

output:

`INDIVISIBLE`

Problem 4: Layer composition

Consider an array of non-negative integers represented as a bar chart. Each array element is represented by a bar whose height equals the value of that element.

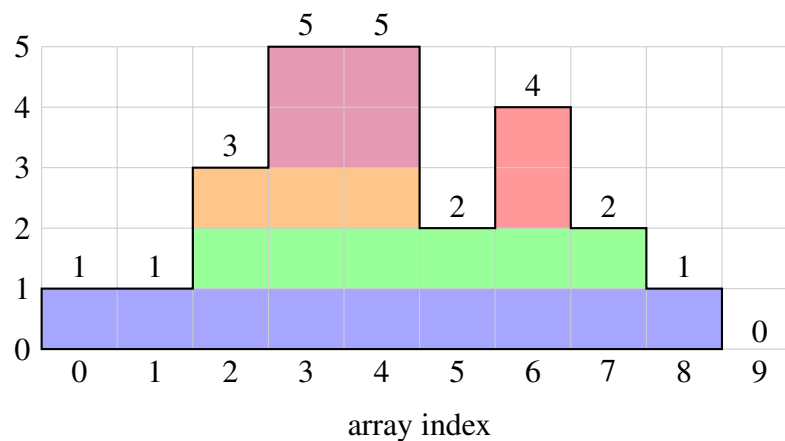


Figure 1: Decomposition of the array $[1, 1, 3, 5, 5, 2, 4, 2, 1, 0]$ into horizontal layers.

We can look upon this chart as a composition of layers. A *layer* is a horizontal rectangle with

maximal size that is described by three integers: *first last delta*.

Such a layer extends from index *first* up to and including index *last*. the value *delta* specifies how much the layer adds to the layers below it, rather than its absolute height in the bar chart.

The input for this problem consists of several lines. The first line is the length of an array, and the remaining lines describe its layers. Note that the order of these lines is not unique. For example, the array [1, 1, 3, 5, 5, 2, 4, 2, 1, 0] (see figure 1) is described by the inputs of the examples 1 and 2 below.

Your program should convert the layers from the input into an array with values. The output of your program must be two lines. The first line contains the length of the array, while the second line contains the array values.

Example 1:

input:

```
10
0 8 1
2 7 1
2 4 1
6 6 2
3 4 2
```

output:

```
10
1 1 3 5 5 2 4 2 1 0
```

Example 2:

input:

```
10
3 4 2
2 4 1
6 6 2
2 7 1
0 8 1
```

output:

```
10
1 1 3 5 5 2 4 2 1 0
```

Example 3:

input:

```
5
2 2 1
1 3 1
```

output:

```
5
0 1 2 1 0
```

Problem 5: Layer decomposition

In this exercise you are required to make a program that is the inverse of the previous problem. Given an array as input, generate its layer representation. Note that the order in which you output the layers is not relevant, since the judge in Themis for this problem accepts any order.

Example 1:

input:

```
10
1 1 3 5 5 2 4 2 1 0
```

output:

```
10
0 8 1
2 7 1
2 4 1
6 6 2
3 4 2
```

Example 2:

input:

```
10
1 1 3 5 5 2 2 2 1 0
```

output:

```
10
3 4 2
2 4 1
2 7 1
0 8 1
```

Example 3:

input:

```
5
1 2 3 2 1
```

output:

```
5
2 2 1
1 3 1
0 4 1
```